Data-Driven Insights from a Real-World Public Report

Title: **DATA ANALYSIS REPORT – Delivery Logistics Dataset**

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Week 1 : Strategic planning and data exploration

Abstract

In this project, we explored a delivery logistics dataset using Python. The dataset contained information about deliveries such as distance, cost, package weight, weather conditions, region, delays, and customer ratings. Our goal was to visually understand how deliveries happen, what affects delivery quality, and what patterns exist behind delays and costs.

Using Python visualizations like histograms, boxplots, scatterplots, heatmaps, and more, we were able to spot clear trends—for example, long-distance deliveries cost more, delays reduce customer ratings, and certain regions experience more issues than others. This project helped us learn how data can reveal real-world operational problems and guide decision-making in the logistics industry.

1. Introduction

Logistics is something we interact with every day—food deliveries, online shopping, courier services, etc. Behind each delivery, there are many factors like distance, traffic, package weight, and weather that affect how smoothly things work.

This project focuses on a delivery logistics dataset to understand these real-world operations. Using Python, we tried to uncover meaningful patterns that could help improve delivery efficiency and customer satisfaction.

2. Need of the Analysis

Delivery operations generate vast amounts of data daily. Without proper analysis, companies miss valuable insights such as:

- What factors increase delivery cost?
- Which regions face the highest delays?
- Do weather and vehicle types impact success rates?
- What influences customer satisfaction (rating)?

Analysing this data helps stakeholders:

- Reduce operational costs
- Optimize route planning
- Improve customer satisfaction

- Predict potential delays
- Enhance decision-making with visual insights

Python visualizations make complex datasets easier to interpret and support strategic decisions in logistics

3. Motivation Behind Choosing the Report / Area

With the rapid increase in e-commerce, efficient delivery systems have become crucial. The motivation behind analysing a delivery logistics dataset stems from:

- Academic interest in real-world data analysis
- Understanding how logistics companies operate
- Relevance to **SDG 9 – Industry, Innovation & Infrastructure**
- Practical importance for businesses that rely on timely deliveries

This dataset inspired the project because it reflects real operational challenges faced by delivery partners—delays, weather issues, cost variations, and regional performance differences.

4. Objective of the Analysis

1. **To analyse distribution patterns** in delivery-related metrics such as distance, weight, cost, and ratings.
2. **To identify relationships** between numeric variables (e.g., distance vs. cost).
3. **To study performance patterns across regions, vehicle types, and weather conditions.**
4. **To derive actionable insights** using Python visualizations that help improve delivery efficiency and customer satisfaction.

5. Report Details

The dataset used originates from a delivery logistics system containing:

- Delivery distance (km)
- Delivery cost
- Package weight
- Vehicle
- Region
- Weather conditions
- Delivery status (delivered / delayed / failed)
- Customer rating

This dataset includes both categorical and numerical attributes, making it suitable for multi-dimensional analysis using Python visual tools.

6. Data Processing Steps

6.1 Data Extraction

- Imported using Python's **Pandas** library and read from a CSV file.
- Imported to PowerBi dashboard

6.2 Data Cleaning

- Created new columns like vehicle_type, (delivered, delayed, failed_delievery) etc using new column option and conditional formatting.
- Changed data type
- Converted numeric columns (rating) to proper categorical columns.

- Removed missing values during plotting.

6.3 Data Modelling

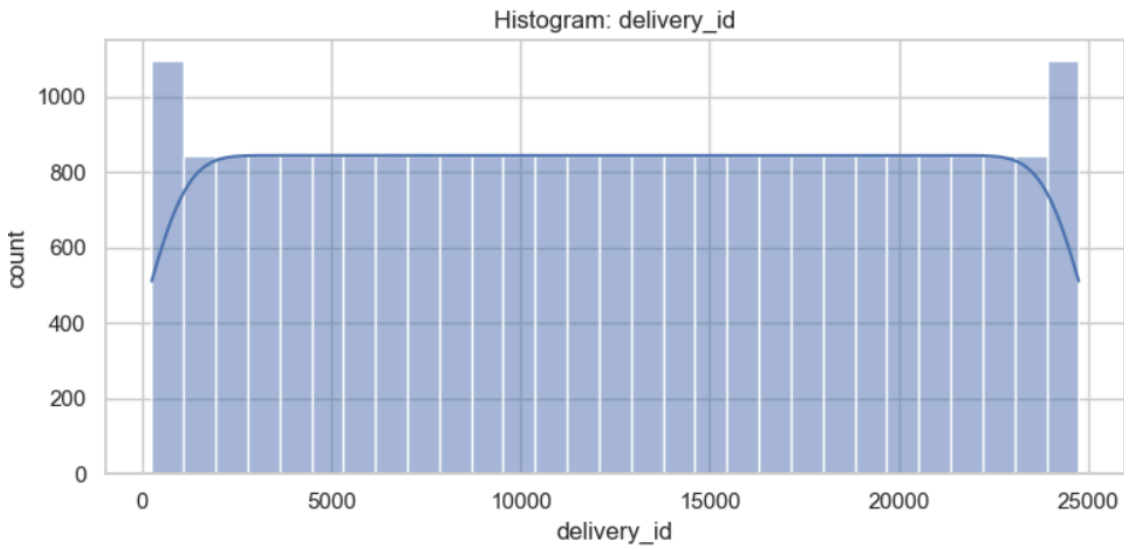
No machine learning model was built; however, clustering labels were already provided. We focused entirely on descriptive analytics.

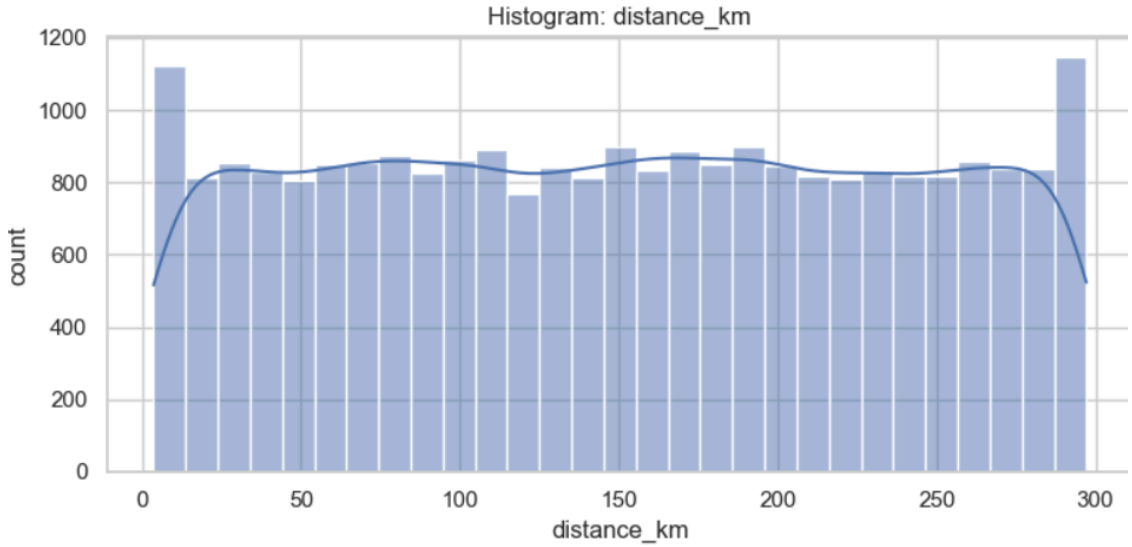
7. Analytical Graphs & Interpretation

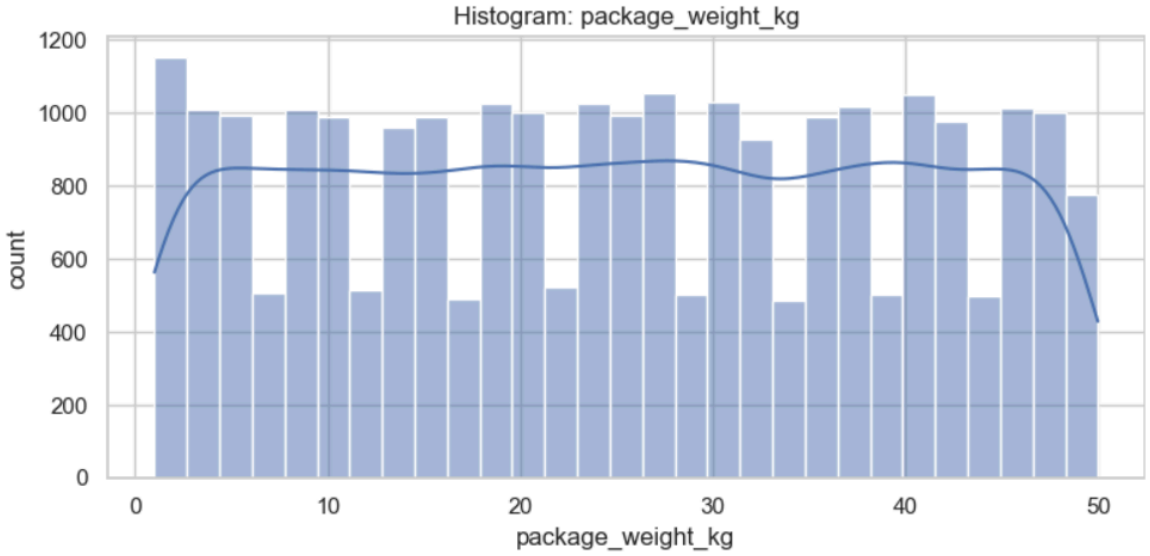
Below is a summary of all plots used and the insights obtained.

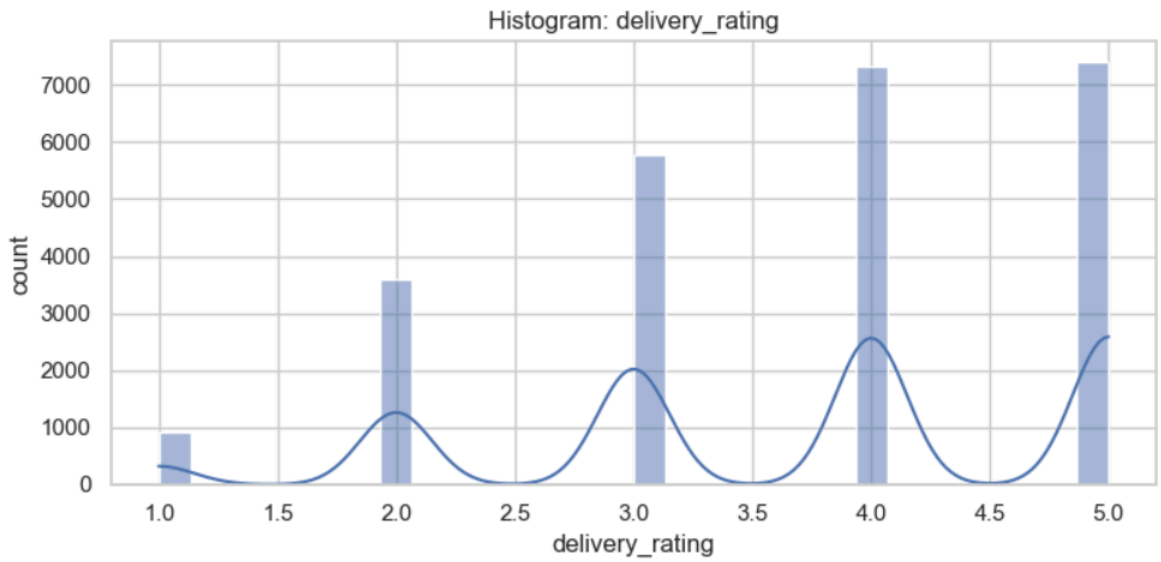
7.1 Histograms (Distribution Plots)

Purpose: Understand how numeric variables are distributed.




Histogram: distance_km
count
distance_km
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300


Histogram: package_weight_kg
count
package_weight_kg
0
200
400
600
800
1000
1200
0
10
20
30
40
50



Columns Used:

- distance_km
- package_weight_kg
- delivery_id
- delivery_rating

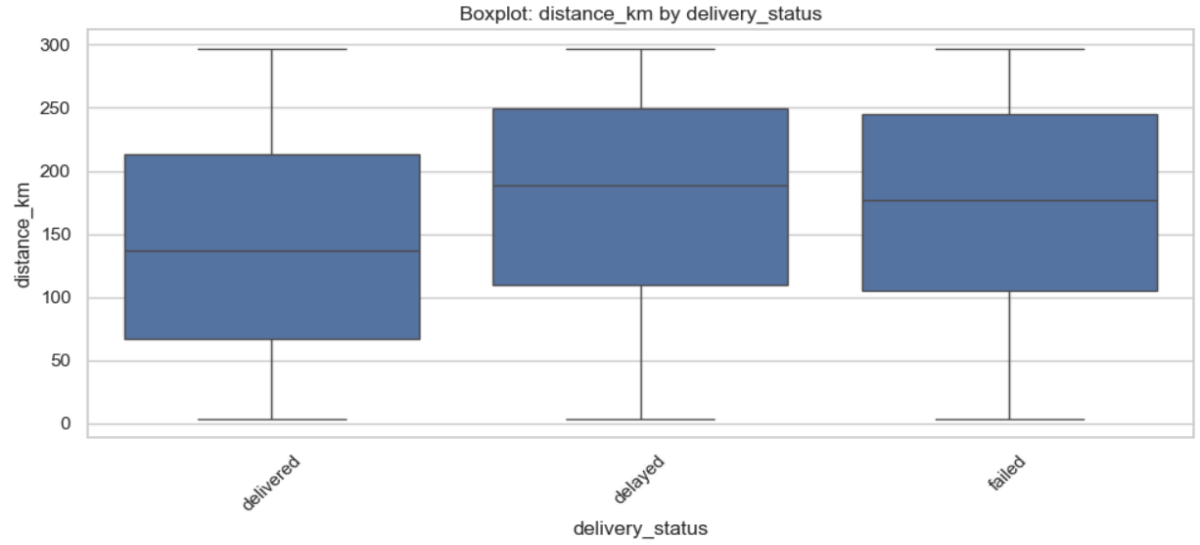
Insights:

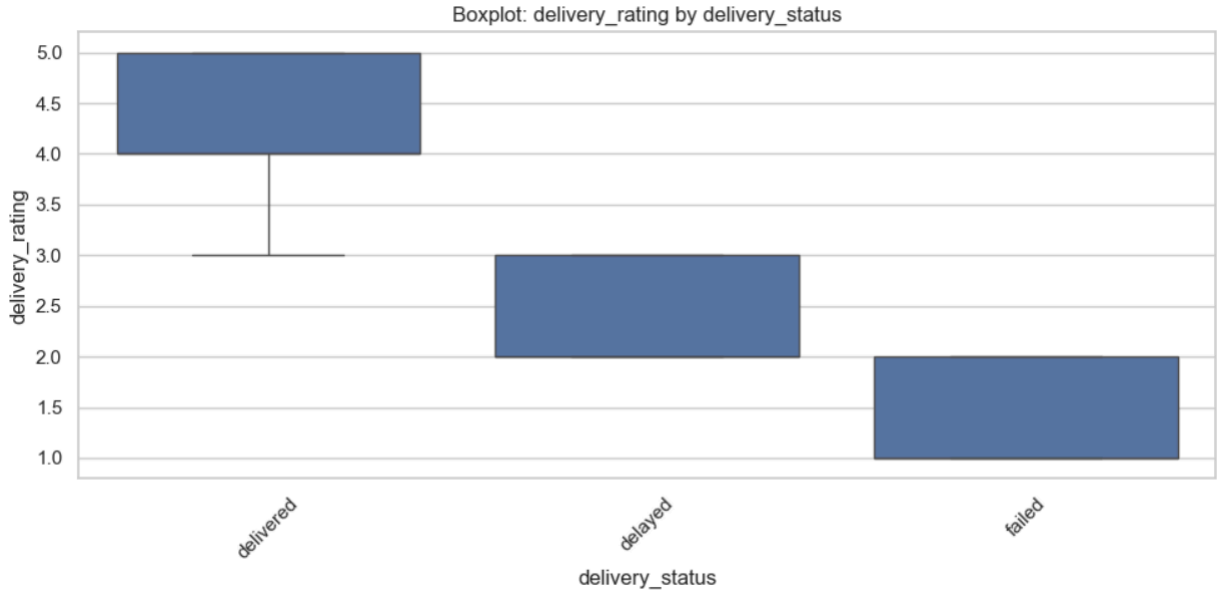
- The distribution of delivery_id values is almost perfectly uniform, indicating that delivery records are evenly spread across the entire ID range (1–25,000).
- This confirms that the dataset does not have missing ID segments or sequencing gaps and that each delivery has an equal likelihood of appearing in any part of the ID range, suggesting a well-generated or well-maintained dataset.
- Delivery distances are **uniformly distributed** across the entire range (0–300 km), meaning deliveries occur equally often at short, medium, and long distances.
- Package weights also show a **fairly uniform distribution**, indicating that light and heavy packages are shipped in almost equal proportions across the dataset.
- The delivery ratings are **highly right-skewed toward positive ratings**, with the majority of customers giving **4-star and 5-star ratings**.
  Lower ratings (1 and 2 stars) occur much less frequently, indicating that **overall**

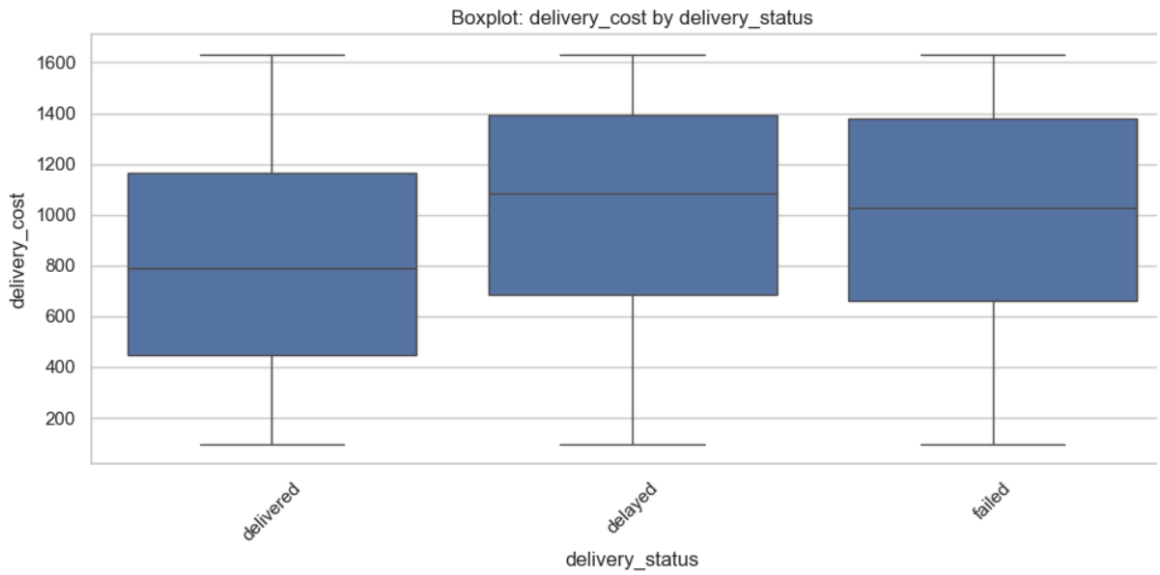
customer satisfaction is high despite some delayed or failed deliveries.

7.2 Boxplots (Category Comparisons)

Purpose: Compare numeric metrics across different categories.







Examples Used:

- distance_km by delivery_status
- delivery_cost by delivery_status
- delivery_rating by delivery_status

Insights:

- **Delayed and failed deliveries tend to occur over longer distances**, as seen by their higher median and upper quartile distances compared to successful (delivered) orders.
- Successful deliveries receive **significantly higher ratings**, while delayed deliveries receive moderate ratings, and failed deliveries get the **lowest ratings**, showing a strong link between delivery outcome and customer satisfaction.
- Delayed and failed deliveries tend to have **higher delivery costs**, suggesting that long-distance or more expensive deliveries are more prone to delays or failure.

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7 Scatterplots (Relationship Between Two Numeric Variables)

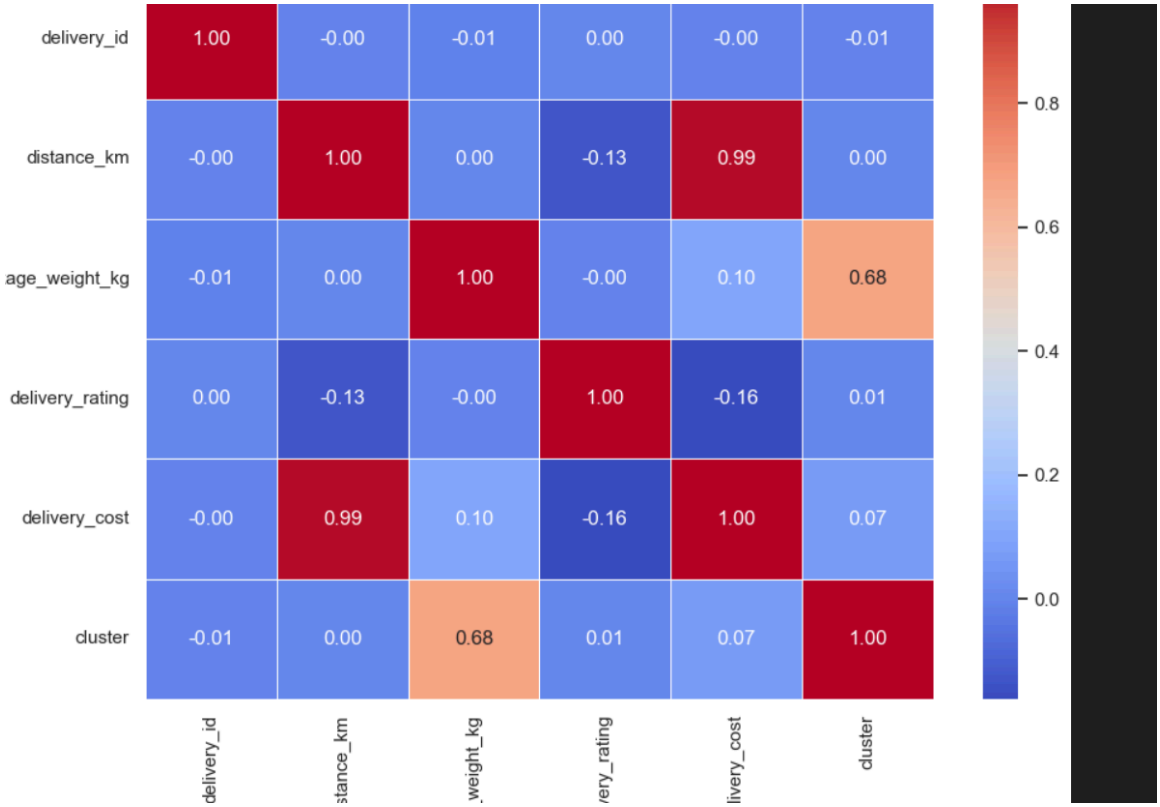
Pairs Used:

- distance_km vs delivery_cost
- package_weight_kg vs delivery_cost
- distance_km vs delivery_rating

Insights:

- **Strong positive correlation** between distance and cost.
- Heavy packages → higher cost.
- Longer distance may slightly reduce ratings.

7.4 Correlation Heatmap (Numeric Relationships)



Purpose: Identify strength and direction of relationships.

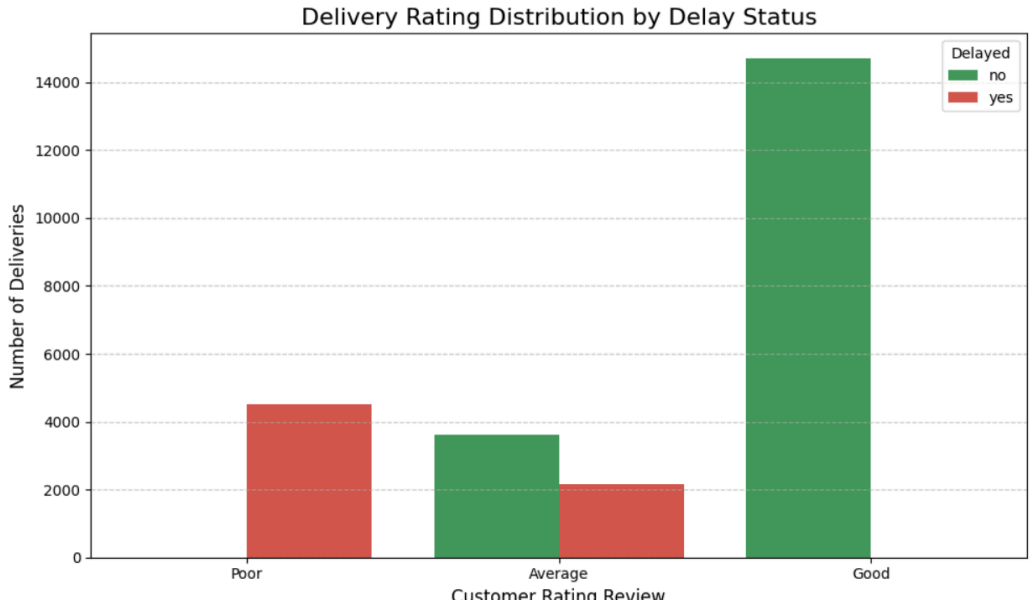
Insights:

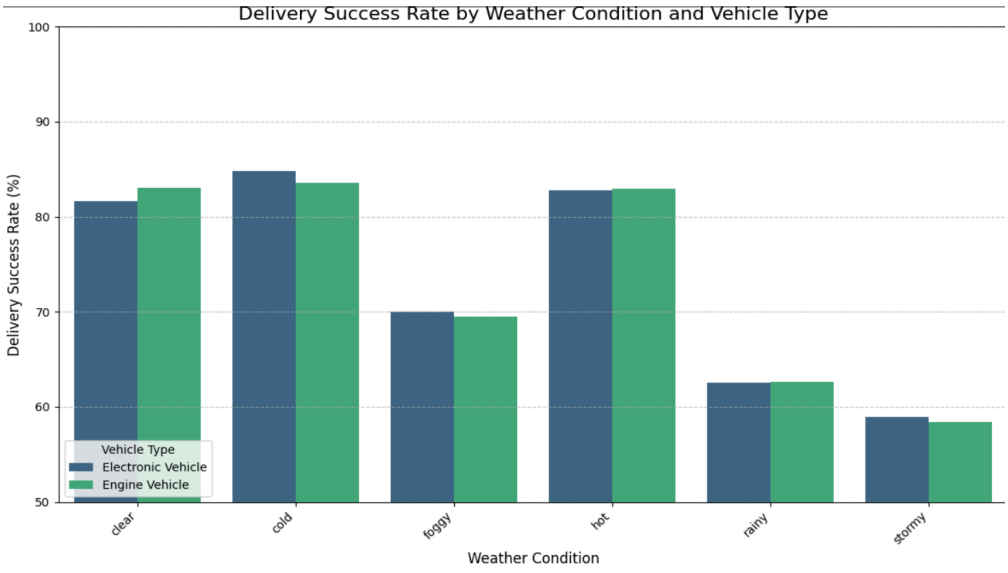
🎬 Distance and delivery cost are almost perfectly correlated (0.99) — longer distances directly drive higher delivery costs.

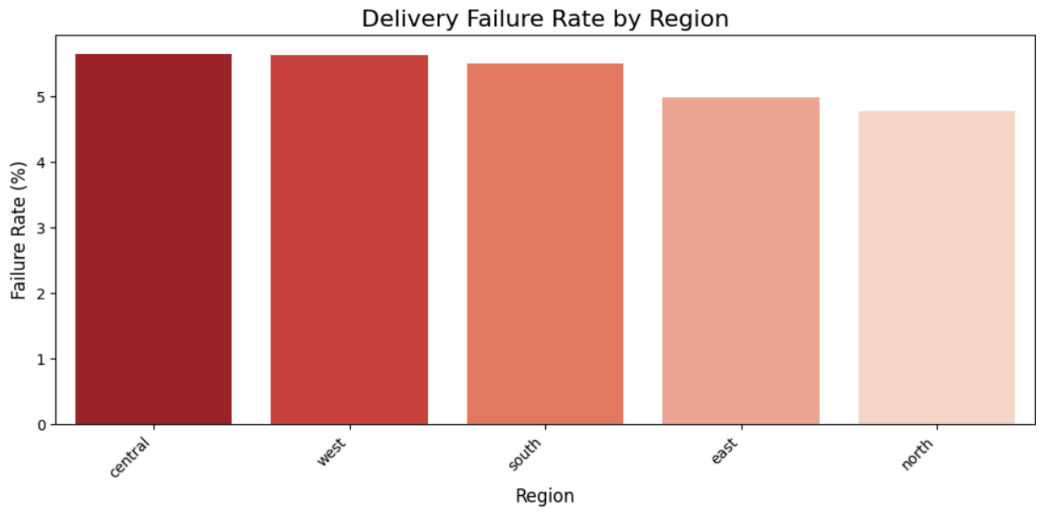
🎬 Cluster is moderately correlated with weight (0.68) — heavier packages tend to fall into a specific cluster.

🎬 Delivery rating has very weak correlations with all other variables — customer ratings are largely independent of distance, cost, or weight.

7.5 Bar Charts







Examples Used:

- Region vs Failure Rate (%)
- Weather Condition vs Delivery success Rate (%)
- Customer Rating Review vs Number of Deliveries

Insights:

- Central Region has the highest failure rate % whereas the least is in the northern region.
- Bad weather conditions (rain, fog) increase delays leading to low Delivery success Rate.
- Deliveries that are not delayed overwhelmingly receive good ratings, while delayed deliveries are far more likely to receive poor or average ratings.

8. Further Scope of Work

A power Bi dashboard



9. Conclusion

This analysis successfully demonstrated how Python-based visualizations can uncover essential insights in a delivery logistics dataset. The study revealed patterns in cost, distance, delays, regional differences, and customer satisfaction. These insights are valuable for improving delivery operations, reducing delays, optimizing resource allocation, and enhancing customer experience.

The project also strengthened our understanding of data-driven decision-making and visualization techniques.

10. Annexure

Include raw data snapshots, DAX formulas, survey forms, or additional supporting documents.

Original-

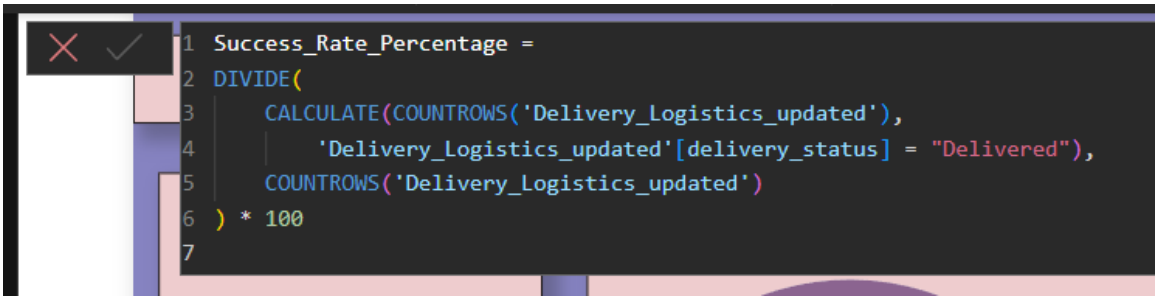
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	delivery_id	delivery_p	package_t	vehicle_ty	delivery_m	region	weather_c	distance_k	package_v	delivery_ti	expected_	delayed	delivery_st	delivery_ra	delivery_cost
2	250.99	delhivery	automobil	bike	same day	west	clear	297	46.96	00:00.0	00:00.0	no	delivered	3	1632.721
3	250.99	xpressbees	cosmetics	ev van	express	central	cold	89.6	47.39	00:00.0	00:00.0	no	delivered	5	640.17
4	250.99	shadowfax	groceries	truck	two day	east	rainy	273.5	26.89	00:00.0	00:00.0	no	delivered	4	1448.17
5	250.99	dhl	electronics	ev van	same day	east	cold	269.7	12.69	00:00.0	00:00.0	no	delivered	3	1486.57
6	250.99	dhl	clothing	van	two day	north	foggy	256.7	37.02	00:00.0	00:00.0	no	delivered	4	1394.56
7	250.99	amazon lo	document	ev bike	express	west	rainy	48.4	33.15	00:00.0	00:00.0	yes	delayed	3	391.45
8	250.99	delhivery	groceries	scooter	same day	central	clear	198.3	43.79	00:00.0	00:00.0	no	delivered	3	1222.87
9	250.99	xpressbees	fragile iten	van	same day	north	cold	114.6	42.63	00:00.0	00:00.0	no	delivered	3	800.89
10	250.99	blue dart	clothing	van	same day	south	hot	142.4	14.06	00:00.0	00:00.0	no	delivered	5	854.18
11	250.99	delhivery	pharmacy	truck	same day	east	foggy	47.1	29.28	00:00.0	00:00.0	no	delivered	5	423.34
12	250.99	fedex	groceries	scooter	same day	east	cold	293.2	22.05	00:00.0	00:00.0	no	delivered	4	1632.15
13	250.99	ecom expr	electronics	truck	two day	east	cold	80.4	46.48	00:00.0	00:00.0	no	delivered	5	541.44
14	250.99	xpressbees	pharmacy	ev van	two day	north	rainy	257.7	20.56	00:00.0	00:00.0	no	delivered	4	1350.18
15	250.99	xpressbees	cosmetics	bike	express	east	stormy	19	3.36	00:00.0	00:00.0	yes	failed	1	155.08
16	250.99	xpressbees	automobil	ev bike	two day	east	stormy	237.9	24.59	00:00.0	00:00.0	no	delivered	3	1263.27
17	250.99	amazon lo	furniture	ev van	express	north	hot	239.3	11.67	00:00.0	00:00.0	yes	delayed	3	1281.51
18	250.99	fedex	document	bike	standard	west	stormy	79	0.73	00:00.0	00:00.0	no	delivered	5	397.19
19	250.99	blue dart	fragile iten	ev bike	standard	central	rainy	170.7	45.62	00:00.0	00:00.0	no	delivered	5	990.36
20	250.99	amazon lo	document	van	same day	north	rainy	90	33.7	00:00.0	00:00.0	no	delivered	3	651.1
21	250.99	fedex	cosmetics	truck	express	north	rainy	64	6.03	00:00.0	00:00.0	yes	failed	2	388.09
22	250.99	fedex	automobil	scooter	two day	east	foggy	185.9	19.1	00:00.0	00:00.0	no	delivered	4	986.8
23	250.99	dhl	electronics	scooter	two day	west	stormy	217.9	14.08	00:00.0	00:00.0	no	delivered	4	1131.74
24	250.99	blue dart	automobil	scooter	same day	north	cold	214.1	40.85	00:00.0	00:00.0	no	delivered	5	1293.05
25	250.99	ecom expr	clothing	bike	standard	central	clear	251.7	34.9	00:00.0	00:00.0	no	delivered	4	1363.2
26	250.99	amazon lo	furniture	van	standard	east	foggy	214.9	5.67	00:00.0	00:00.0	no	delivered	5	1091.51
27	250.99	shadowfax	document	ev van	two day	north	stormy	115.4	0.67	00:00.0	00:00.0	no	delivered	4	578.59
28	250.99	dhl	clothing	ev bike	express	east	foggy	293.8	41.03	00:00.0	00:00.0	yes	delayed	3	1632.721
29	250.99	amazon lo	cosmetics	scooter	standard	east	clear	286.8	46.67	00:00.0	00:00.0	no	delivered	3	1574.01

Updated-

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	delivery_id	delivery_p	package_t	vehicle	vehicle_ty	delivery_m	region	weather_c	distance_k	package_v	Order_Date	Delivery_Date	delivery_st	delayed	Delievered	Failed_Del	delivery_ra	Rating_Re	delivery_c	Month Name		
2	251	delhivery	automobil	bike	Engine Vel	same day	west	clear	297	47	16-02-2025	20-02-2025	delivered	no	yes	no	3	Average	1632.721	February		
3	251	xpressbee	cosmetics	ev van	Electronic	express	central	cold	89.6	47	13-03-2025	24-03-2025	delivered	no	yes	no	5	Good	640.17	March		
4	251	shadowfa	groceries	truck	Engine Vel	two day	east	rainy	273.5	27	20-11-2025	29-11-2025	delivered	no	yes	no	4	Good	1448.17	November		
5	251	dhl	electronic	ev van	Electronic	same day	east	cold	269.7	13	08-04-2025	21-04-2025	delivered	no	yes	no	3	Average	1486.57	April		
6	251	dhl	clothing	van	Engine Vel	two day	north	foggy	256.7	37	11-09-2025	21-09-2025	delivered	no	yes	no	4	Good	1394.56	September		
7	251	amazon lo	document	ev bike	Electronic	express	west	rainy	48.4	33	20-08-2025	30-08-2025	delayed	yes	no	no	3	Average	391.45	August		
8	251	delhivery	groceries	scooter	Engine Vel	same day	central	clear	198.3	44	12-05-2025	19-05-2025	delivered	no	yes	no	3	Average	1222.87	May		
9	251	xpressbee	fragile iten	van	Engine Vel	same day	north	cold	114.6	43	17-04-2025	26-04-2025	delivered	no	yes	no	3	Average	800.89	April		
10	251	blue dart	clothing	van	Engine Vel	same day	south	hot	142.4	14	17-04-2025	24-04-2025	delivered	no	yes	no	5	Good	854.18	April		
11	251	delhivery	pharmacy	truck	Engine Vel	same day	east	foggy	47.1	29	21-05-2025	31-05-2025	delivered	no	yes	no	5	Good	423.34	May		
12	251	fedex	groceries	scooter	Engine Vel	same day	east	cold	293.2	22	01-08-2025	09-08-2025	delivered	no	yes	no	4	Good	1632.15	August		
13	251	ecom expr	electronic	truck	Engine Vel	two day	east	cold	80.4	46	30-11-2025	07-12-2025	delivered	no	yes	no	5	Good	541.44	November		
14	251	xpressbee	pharmacy	ev van	Electronic	two day	north	rainy	257.7	21	06-06-2025	17-06-2025	delivered	no	yes	no	4	Good	1350.18	June		
15	251	xpressbee	cosmetics	bike	Engine Vel	express	east	stormy	19	3	05-12-2025	15-12-2025	failed	yes	no	yes	1	Poor	155.08	December		
16	251	xpressbee	automobil	ev bike	Electronic	two day	east	stormy	237.9	25	10-02-2025	19-02-2025	delivered	no	yes	no	3	Average	1263.27	February		
17	251	amazon lo	furniture	ev van	Electronic	express	north	hot	239.3	12	01-01-2025	10-01-2025	delayed	yes	no	no	3	Average	1281.51	January		
18	251	fedex	document	bike	Engine Vel	standard	west	stormy	79	1	08-09-2025	12-09-2025	delivered	no	yes	no	5	Good	397.19	September		
19	251	blue dart	fragile iten	ev bike	Electronic	standard	central	rainy	170.7	46	31-01-2025	09-02-2025	delivered	no	yes	no	5	Good	990.36	January		
20	251	amazon lo	document	van	Engine Vel	same day	north	rainy	90	34	29-06-2025	08-07-2025	delivered	no	yes	no	3	Average	651.1	June		
21	251	fedex	cosmetics	truck	Engine Vel	express	north	rainy	64	6	23-12-2025	03-01-2026	failed	yes	no	yes	2	Poor	388.09	December		
22	251	fedex	automobil	scooter	Engine Vel	two day	east	foggy	185.9	19	03-07-2025	14-07-2025	delivered	no	yes	no	4	Good	986.8	July		
23	251	dhl	electronic	scooter	Engine Vel	two day	west	stormy	217.9	14	13-09-2025	25-09-2025	delivered	no	yes	no	4	Good	1131.74	September		
24	251	blue dart	automobil	scooter	Engine Vel	same day	north	cold	214.1	41	13-06-2025	21-06-2025	delivered	no	yes	no	5	Good	1293.05	June		
25	251	ecom expr	clothing	bike	Engine Vel	standard	central	clear	251.7	35	05-04-2025	13-04-2025	delivered	no	yes	no	4	Good	1363.2	April		
26	251	amazon lo	furniture	van	Engine Vel	standard	east	foggy	214.9	6	22-09-2025	29-09-2025	delivered	no	yes	no	5	Good	1091.51	September		

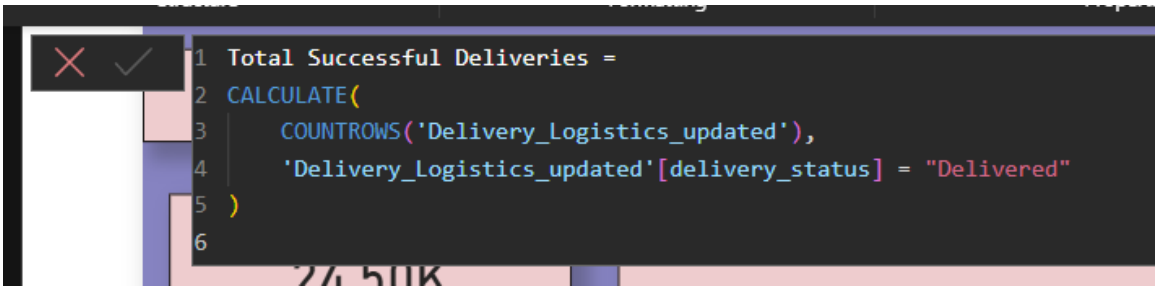
DAX formulae –

```
Success Rate =
VAR Total = COUNTROWS('Delivery_Logistics_updated')
VAR Success =
    CALCULATE(
        COUNTROWS('Delivery_Logistics_updated'),
        'Delivery_Logistics_updated'[delivery_status] = "Delivered"
    )
RETURN
DIVIDE(Success, Total)
```



```
Success_Rate_Percentage =
DIVIDE(
    CALCULATE(COUNTROWS('Delivery_Logistics_updated'),
        'Delivery_Logistics_updated'[delivery_status] = "Delivered"),
    COUNTROWS('Delivery_Logistics_updated')
) * 100

```



```
Total Successful Deliveries =
CALCULATE(
    COUNTROWS('Delivery_Logistics_updated'),
    'Delivery_Logistics_updated'[delivery_status] = "Delivered"
)

```